

CHUQUR O'RGANISH USULLARI · Amaliy mashg'ulot (Practice)

4-mavzu. Rekurrent neyron tarmoqlar: RNN, LSTM va GRU

2026–2027 o'quv yili · 30 ta topshiriq

Muhitni sozlash. Barcha kodlar Python 3.11/3.12 (CPU) va quyidagi qat'iy versiyalar bilan sinovdan o'tgan (requirements.txt); ma'lumotlar internetsiz: sintetik yoki topshiriq matnida berilgan.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
# numpy==2.4.4 scipy==1.17.1 pandas==3.0.2 scikit-learn==1.8.0 scikit-image==0.26.0
# matplotlib==3.10.9 pillow==12.2.0 opencv-python-headless==4.13.0.92 torch==2.14.0 torchvision==0.29.0
```

Hujjatlar: <https://docs.pytorch.org/docs/stable/nn.html#recurrent-layers>, https://docs.pytorch.org/docs/stable/generated/torch.nn.utils.rnn.pack_padded_sequence.html, https://docs.pytorch.org/docs/stable/generated/torch.nn.utils.clip_grad_norm_.html.

Mavzu. Ketma-ketliklarni modellash: RNN, BPTT, vanishing/exploding gradient, LSTM, GRU, padding va packing, Bahdanau attention, beam search, til modeli va perplexity.

Maqsadlar. RNN/LSTM/GRU qadamini qo'lda va kodda hisoblash; BPTT gradientini chiqarish va sonli tekshirish; gradient so'nishini o'lchash; parametrlarni sanash; kichik korpusda char-level LSTM va vaqt qatori modelini o'qitib, bazaviy usul bilan taqqoslash.

Baholash (100 ball). 1–16: har biri 2 ball (32); 17–28: har biri 4 ball (48); 29, 30: har biri 10 ball (20). Kod topshiriqlarida: to'g'ri ishlashi 60%, talab qilingan chiqish formati 20%, natija talqini 20%. Umumiy talab: har bir kod topshirig'i alohida funksiya; `torch.manual_seed(42)`, `np.random.default_rng(42)`, CPU; sonlar 4 xonagacha yaxlitlanadi.

A. Qo'lda hisoblash, isbot va tushunchalar (1–16)

- Padding.** Uzunliklari 4, 1, 6 bo'lgan, $d = 5$ o'lchamli uchta ketma-ketlik `batch_first=True` bilan to'ldiriladi. Tenzor shakli, jami va haqiqiy pozitsiyalar soni, to'ldiruvchi ulushi hamda `pack_padded_sequence` beradigan `batch_sizes` ro'yxatini toping.
- Skalyar RNN.** $h_t = \tanh(0,8x_t - 0,6h_{t-1} + 0,1)$, $h_0 = 0$, $x = (1; 2; -1)$. h_1, h_2, h_3 ni toping (4 xona).
- Vektor RNN.** $W_{xh} = \begin{pmatrix} 0,5 & -1 \\ 1 & 0,5 \end{pmatrix}$, $W_{hh} = \begin{pmatrix} 0,5 & 0 \\ 0 & -0,5 \end{pmatrix}$, $\mathbf{b} = \mathbf{0}$, $h_0 = \mathbf{0}$, $x_1 = (1; 0)$, $x_2 = (0; 1)$. h_1, h_2 ni, so'ng $W_{hy} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, $\mathbf{b}_y = \mathbf{0}$ uchun $o_2 = W_{hy}h_2$ va `softmax( $o_2$ )` ni hisoblang.
- Parametrlar soni (PyTorch hisobi, ikki siljish vektori bilan).** a) `nn.RNN(16, 32, num_layers=2)`; b) `nn.LSTM(16, 32)`; c) `nn.GRU(16, 32, bidirectional=True)`.
- BPTT (skalyar).** $h_t = \tanh(w_{xh}x_t + w_{hh}h_{t-1} + b)$, $w_{xh} = 0,5$, $w_{hh} = 1$, $b = 0$, $h_0 = 0$, $x = (2; 1)$, $L = \frac{1}{2}h_2^2$. L , $\partial L / \partial w_{hh}$, $\partial L / \partial w_{xh}$, $\partial L / \partial b$ ni toping.
- Isbot.** $\gamma = \sup |\varphi'|$, $s = \|W_{hh}\|_2$ bo'lsa, $\|\partial \mathbf{h}_T / \partial \mathbf{h}_t\|_2 \leq (\gamma s)^{T-t}$ ekanini isbotlang. $\tanh$, $s = 0,8$ da $T - t = 20$ uchun chegarani va chegara 10^{-3} dan kichik bo'ladigan eng kichik $T - t$ ni toping.
- Gradient clipping.** $\mathbf{g} = (2; -1; 2)$, $\tau = 1,5$. Norma bo'yicha kesish natijasini toping. Qiymat bo'yicha kesish (`clip_grad_value_, 1,5`) natijasini ham toping; qaysi biri yo'nalishni saqlaydi?
- LSTM qadami.** Skalyar LSTM: $x_1 = 2$, $h_0 = 0$, $c_0 = -0,5$, $a_f = 0,5x$, $a_i = x - 1$, $a_c = -0,5x$, $a_o = x$. $f_1, i_1, \tilde{c}_1, o_1, c_1, h_1$ ni toping.
- Qo'shuvchi xotira yo'li.** Bevosita yo'l bo'yicha $\partial \mathbf{c}_T / \partial \mathbf{c}_t = \text{diag}(\prod_{k=t+1}^T \mathbf{f}_k)$ ekanini ko'rsating. $f_k \equiv \sigma(0), \sigma(2), \sigma(4)$ bo'lganda $T - t = 30$ uchun qiymatlarni hisoblang.
- Forget gate bias.** $W_f \mathbf{z} = 0$ deb, $b_f = 1$ va $b_f = 2$ uchun xotira yarmiga tushadigan qadamlar soni $n = \ln 0,5 / \ln f$ ni toping. Yarim yemirilish 50 qadam bo'lishi uchun f va b_f qanday bo'lishi kerak?
- GRU qadami (PyTorch konvensiyasi).** $x = 1$, $h_0 = -0,5$, $r = \sigma(x + h_0)$, $u = \sigma(0,5x - h_0)$, $\tilde{h} = \tanh(x + r \cdot 2h_0)$, $h_1 = (1 - u)\tilde{h} + uh_0$. Barcha oraliq qiymatlarni toping.
- GRU xossalari.** a) $|h_{t-1}| < 1$ bo'lsa, $|h_t| < 1$ ekanini isbotlang. b) Yakobian $\partial \mathbf{h}_t / \partial \mathbf{h}_{t-1}$ formulasini chiqaring va $\mathbf{u}_t \rightarrow \mathbf{1}$ da nima bo'lishini tushuntiring.
- Additive attention.** $\mathbf{s} = (1; 0)$, $\mathbf{h}_1 = (1; 1)$, $\mathbf{h}_2 = (0; -1)$, $\mathbf{h}_3 = (-1; 0,5)$, $W = U = I$, $\mathbf{v} =$

$(1; -1)$: $e_t = \mathbf{v}^T \tanh(W\mathbf{s} + U\mathbf{h}_t)$. $e_t, \alpha_t = \text{softmax}(e)_t$ va $\mathbf{c} = \sum \alpha_t \mathbf{h}_t$ ni toping.

14. **Beam search.** $p(y_1)$: $A 0,6, B 0,4$; $p(y_2|A)$: $A 0,3, B 0,4, C 0,3$; $p(y_2|B)$: $A 0,8, B 0,1, C 0,1$. Greedy va $k = 2$ beam natijalari, ehtimollari va log-ehtimollarini toping.
15. **Perplexity va LM hajmi.** a) To'g'ri tokenlar ehtimollari 0,5; 0,2; 0,4; 0,1; 0,25: CE (nat) va PPL? $V = 40$ tekis model PPL si? b) So'z LM: $V = 8000, E = H = 256$, bitta LSTM, Linear(H, V). Jami parametrlar va weight tying (embedding va chiqish og'irligi umumiy) dan keyingi son.
16. **Tushunchalar** (har biriga 2–3 jumla). a) Nega bidirectional RNN ni real vaqtli vaqt qatori bashoratida ishlatib bo'lmaydi, NER da esa bo'ladi? b) Vaqt qatorini o'quv/test ga nega aralashtirmasdan bo'lish kerak? c) Teacher forcing va exposure bias. d) Truncated BPTT da `h.detach()` nima qiladi va qanday cheklov kiritadi?

B. Dasturlash (17–30)

Har bir funksiya natijani `[T<nomer>]` prefiksi bilan chop etadi; hammasini `main()` ishga tushiradi. Tekshirishlarda `float64 (.double())` ishlatilg.

17. **NumPy RNN.** `rnn_forward(X, Wxh, Whh, b, h0) -> Hs` ($X: (T, d), Hs: (T, H)$). `torch.manual_seed(42); rng=np.random.default_rng(42); X=rng.normal(size=(5,3)); rnn=nn.RNN(3,4).double()` og'irliklarini oling ($b = b_{ih} + b_{hh}$). h_T ni va `nn.RNN` bilan maksimal farqni chiqaring.
18. **BPTT + gradient check.** `bptt_grad(X, Wxh, Whh, b) -> (dWhh, dWxh)`, $L = \sum_i h_{T,i}$. `rng=default_rng(42); Wxh=rng.normal(0,.5,(4,3)), Whh=rng.normal(0,.5,(4,4)), X=rng.normal(size=(6,3))` (shu tartibda), $b = 0$. $\partial L / \partial W_{hh}[1, 2]$ ni markaziy ayirma ($\varepsilon = 10^{-6}$) va autograd bilan solishtiring; $\|dW_{hh}\|_F, \|dW_{xh}\|_F$ ni chiqaring.
19. **Parametrlarni sanash.** `count_params(m)`; 4- va 15b-topshiriq javoblarini PyTorch bilan tekshiring (tying: `out.weight = emb.weight`).
20. **LSTM yacheykasi.** `lstm_step(x, h, c, W_ih, W_hh, b_ih, b_hh) -> (h, c)`, darvozalar tartibi `[i, f, g, o]`. `torch.manual_seed(42), cell=nn.LSTMCell(3,5).double(), so'ng rng=default_rng(42): x=rng.normal(size=3), h=rng.normal(size=5), c=rng.normal(size=5). c1` va `nn.LSTMCell` bilan farqni chiqaring.
21. **GRU yacheykasi.** `gru_step(x, h, W_ih, W_hh, b_ih, b_hh)`, tartib `[r, z, n]`; 17/20 dagi kabi `nn.GRUCell(3,5)`, `x, h`. So'ng `GRUCell(1,1)` og'irliklarini `weight_ih=[1, 0.5, 1], weight_hh=[1, -1, 2]`, `bias = 0` qilib 11-topshiriq javobini tekshiring.
22. **Spektral radius va so'nish.** `jac_norm(rho, T=60, H=16): g=torch.Generator().manual_seed(42); float64 da W=randn(16,16), Wx=randn(16,1), x=randn(60,1)` (shu tartibda, `generator=g`); $W *= \rho / \max|\text{eig}(W)|$; $h_t = \tanh(W h_{t-1} + W_x x_t)$, $h_0 = 0$. `torch.autograd.functional.jacobian` bilan $\|\partial h_{60} / \partial h_0\|_2$ ni $\rho \in \{0,5; 0,9; 1; 1,5; 3\}$ uchun chiqaring va natijani 6-topshiriq bilan bog'lang.
23. **Samarali ufq.** `effective_horizon(cell, T=100, d=4, tol=1e-3): x=torch.randn(1,T,d)` (`torch.manual_seed(0)` dan keyin), $L = \sum h_T$, `reach(t) = ||∂L/∂x_t||`; ufq — `reach(t) > 10-3 maxreach` bo'lgan qadamlar soni. Modellar (`torch.manual_seed(42)` dan keyin, $H = 32$, `batch_first=True`): RNN, GRU, LSTM, LSTM (b_{ih} va b_{hh} ning `[32:64]` qismi 1,0).
24. **Clipping.** `p=torch.zeros(3, requires_grad=True), p.grad=tensor([2., -1., 2.]): clip_grad_norm_(max_norm=1.5)` (qaytgan qiymat, yangi **g**, yangi norma) va `clip_grad_value_(1.5)` (**g** va eski yo'nalish bilan cos). 7-topshiriqni tasdiqlang.
25. **Packing.** `torch.manual_seed(42); seqs=[torch.randn(L,3) for L in (5,2,7)], so'ng gru=nn.GRU(3,4,batch_first=True).` `pad_sequence + pack_padded_sequence(enforce_sorted=False)`. Oxirgi holatni har bir ketma-ketlikni alohida o'tkazish bilan solishtiring: packed va to'ldirilgan (packsiz) variant uchun maksimal farqlar, `batch_sizes`.
26. **Attention funksiyasi.** `additive_attention(s, Hs, W, U, v) -> (e, alpha, c)` (`softmax`'ni `e - e.max()` bilan barqaror yozing); 13-topshiriqni tekshiring.

- 27. Beam search.** Lug'at $\{a, b, c\}$, bigram jadval $p(y_t|y_{t-1})$ (qatorlar: oldingi token; ustunlar a, b, c):
 $\langle s \rangle$: 0,05; 0,50; 0,45; a : 0,10; 0,25; 0,65; b : 0,35; 0,45; 0,20; c : 0,40; 0,10; 0,50. `beam_search(k, length=3)` $\rightarrow$ (`satr`, `logp`) (har qadamda kV nomzoddan eng yaxshi k). $k = 1, 2, 3$ va to'liq qidiruv (27 ketma-ketlik) natijalarini chiqaring.
- 28. Vaqt qatori (many-to-one).** `torch.manual_seed(42)`; `t=torch.arange(0,200,0.1)`, $s = \sin t + 0,5 \sin 0,3t$; oyna $W = 30$: $X_i = s_{i:i+30}$, $y_i = s_{i+30}$; birinchi 1400 juftlik — train, qolgani — test. `LSTM(1,16)+Linear(16,1)` (oxirgi qadamdan), Adam `lr=1e-2`, 15 epoxa, `randperm(1400).split(50)`, MSE, `clip_grad_norm_(1.0)`. Test MSE ni "oxirgi qiymatni takrorla" bazaviy usuli bilan solishtiring.
- 29. Ishorani eslab qolish.** `g=torch.Generator().manual_seed(42)`; `make_sign_data(n, T, g)`: `s=randint(0,2,(n,))`, `generator=g`)*2-1, `x=randn(n,T,1,generator=g)`, `x[:,0,0]=s`, nishon (`s>0`). Train $n = 1024$, so'ng test $n = 512$, $T = 30$. Model: `LSTM(1,16)+Linear(16,2)`, `bias_ih` va `bias_hh` ning [16:32] qismi $b_f/2$; $b_f \in \{0, 2\}$ (har biri oldidan `torch.manual_seed(42)`). Adam `1e-2`, 20 epoxa, batch 64 (`randperm`), cross-entropy, clip 1,0. Test aniqligini chiqaring va izohlang.
- 30. Char-level LSTM til modeli.** Korpus (aynan shu satr, 90% train / oxirgi 10% val):

```
TEXT = ("bilim olish har bir inson uchun muhim. kitob o'qish bilimni oshiradi. "
        "yaxshi kitob yaxshi do'st. har kuni kitob o'qing va yangi so'zlarni o'rganing. "
        "ilm olish uchun sabr kerak. sabr qilgan inson maqsadiga yetadi. "
        "bilim kuch, kitob esa bilim manbai. har bir kitob yangi dunyo. "
        "yaxshi do'st va yaxshi kitob insonni yaxshi yo'lga boshlaydi. "
        "ilm olish uchun har kuni o'qing, yozing va o'ylang. ")
```

Lug'at `sorted(set(TEXT))`. CharLM: `Embedding(V,16)`, `LSTM(16,64)`, `Linear(64,V)`. `torch.manual_seed(42)`; Adam `1e-2`; 400 qadam; har qadamda `torch.randint(0, len(tr)-33, (32,))`, `generator=g` (g — seed 42) bilan 32 ta 32-belgili oyna; clip 1,0. Har 50 qadamda train va val PPL; eng yaxshi val PPL holatini saqlang (early stopping) va undan "kitob " dan boshlab 60 belgi greedy generatsiya qiling. Tekis model PPL ($= V$) bilan taqqoslang va overfitting'ni izohlang.